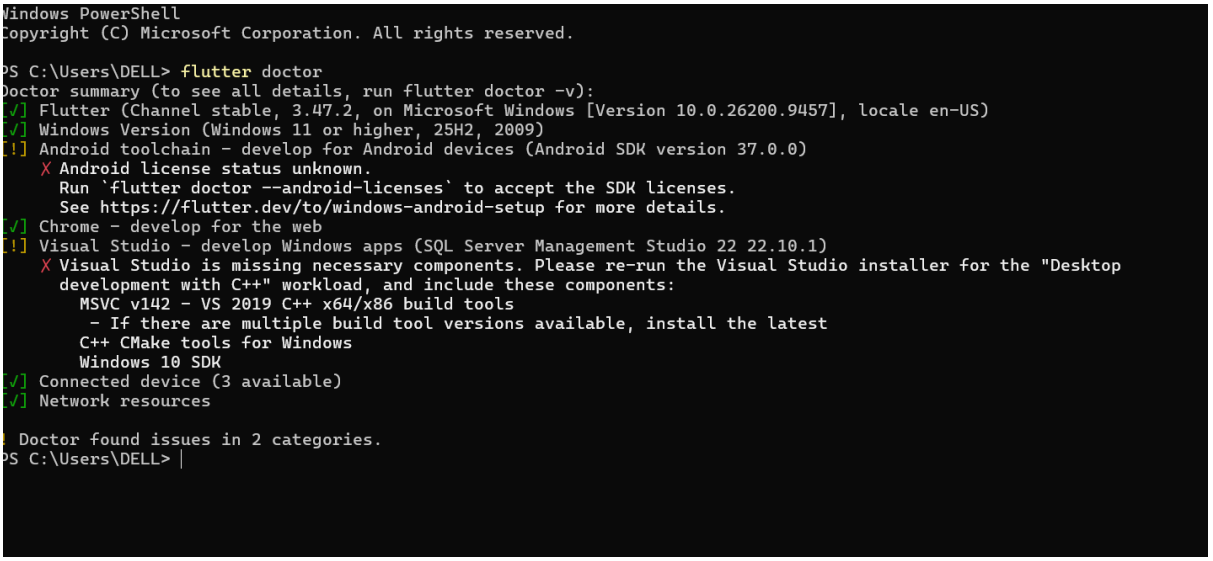
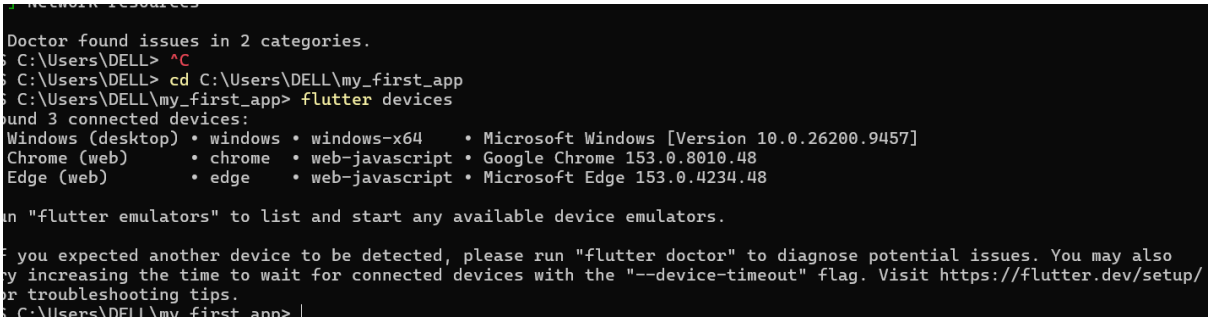
Task 1: Flutter & Dart Environment Setup

1. Open your terminal, run `flutter doctor` to verify your installation, and paste the output summary into your document.



2. Initialize a new Flutter application named `my_first_app` using the terminal, navigate into the project folder, and list all available target devices using `flutter devices`.



3. Answer the following conceptual question:
- What is the difference between 'Hot Reload' and 'Hot Restart' in Flutter? Explain when a developer should use each.

Use Hot Reload for most UI work: tweaking layouts, colors, text, padding, or widget structure inside `build()` methods, where you want to see changes instantly without losing your place in the app.

Use Hot Restart when a change can't be applied to the live app state, such as:

- changes to `main()`, global variables, or static fields
- changes to initializers of state (e.g. `initState()` logic)
- changing an enum or converting a `StatelessWidget` to a `StatefulWidget`
- when the app gets into a confusing state and you want a clean start without a full rebuild

Task 2: MySQL Database Management

1. Log into your local MySQL server as root via the terminal/command line and execute SQL commands to:
- Create a database named `school`.
- Create a `students` table with fields: `id` (AUTO_INCREMENT PRIMARY KEY), `name` (VARCHAR 100), `email` (VARCHAR 150 UNIQUE), and `enrolled_on` (DATE).
- Insert two sample student records.
- Run a query to retrieve all records from the `students` table.

2. Paste the SQL script used into your document alongside a screenshot or terminal output showing the results of your `SELECT *` query.

3. Security Reflection: Explain why using the `root` database account for direct application backend connections is poor security practice, and detail the SQL commands required to create a dedicated, least-privileged application user.

```sql
CREATE TABLE students (
    id INT AUTO_INCREMENT PRIMARY KEY,
    name VARCHAR(100),
    email VARCHAR(150) UNIQUE,
    enrolled_on DATE
);

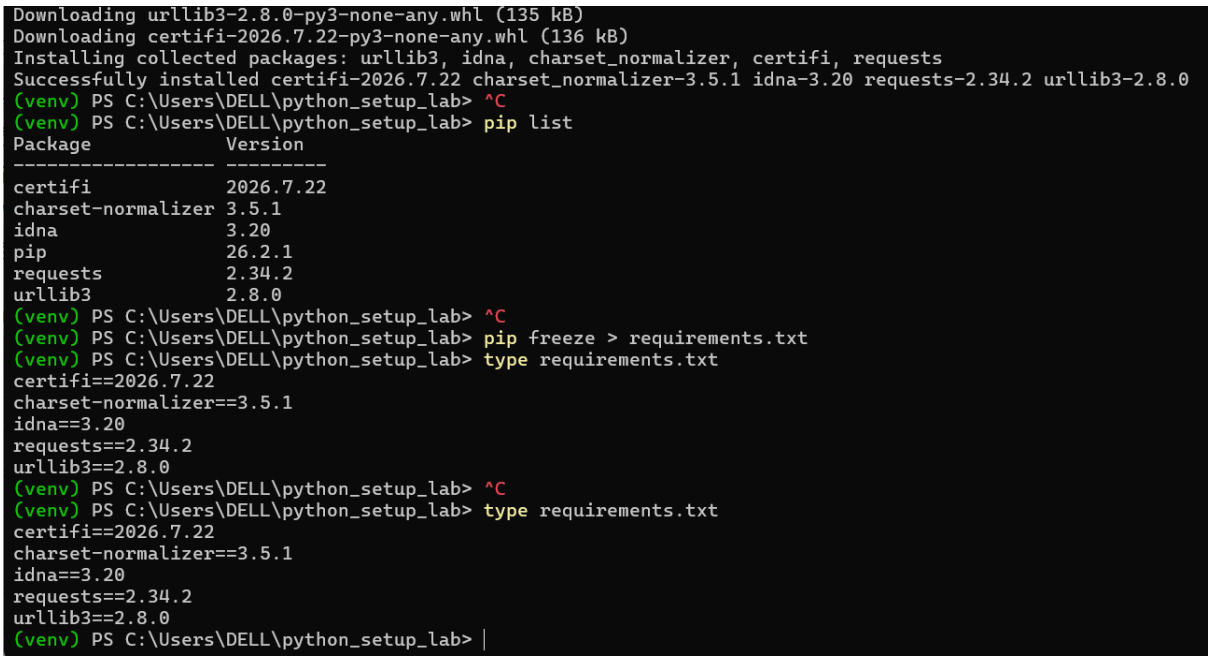
INSERT INTO students (name, email, enrolled_on) VALUES
('Hassan Noor', 'hassan@125.com', '2026-01-15'),
('Amina Mohamed', 'amina@2570.com', '2026-01-20');
```

```
SELECT * FROM students;
```

```
+----+---------------+-----------------+-------------+
| id | name          | email           | enrolled_on |
+----+---------------+-----------------+-------------+
|  1 | Hassan Noor   | hassan@125.com  | 2026-01-15  |
|  2 | Amina Mohamed | amina@2570.com  | 2026-01-20  |
+----+---------------+-----------------+-------------+
```

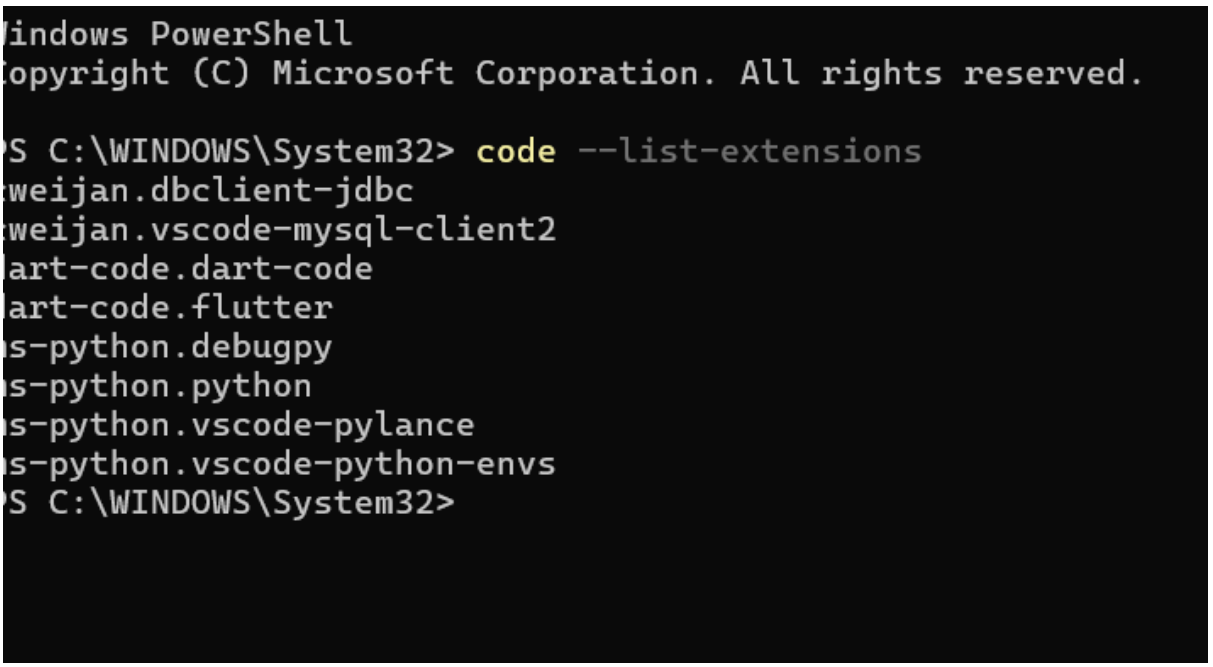
ecurity practice
n's credentials are
and user accounts. A
s safer and follows the
principle of least privilege.

Task 3: Python Environment & Dependency Isolation



Task 4: VS Code Workspace Configuration

1. Install the required extensions in VS Code: `Flutter`, `Dart`, `Python`, `Pylance`, and `MySQL`.



2. Configure VS Code to select and use the Python interpreter located inside your `python_setup_lab/venv` directory.
3. Take a single, full-screen screenshot of your VS Code window displaying:
   - The active Extensions panel showing installed plugins.
   - The integrated terminal open with the `(venv)` prompt active.
   - The bottom status bar showing the selected Python interprete

